

FIRST NOTIONS

OF

LOGIC

(PREPARATORY TO THE STUDY OF GEOMETRY)

BY

AUGUSTUS DE MORGAN,

OF TRINITY COLLEGE, CAMBRIDGE,

PROFESSOR OF MATHEMATICS IN UNIVERSITY COLLEGE, LONDON.

The root of all the mischief in the sciences, is this; that falsely magnifying and admiring the powers of the mind, we seek not its real helps.—BACON.

LONDON :

PRINTED FOR TAYLOR AND WALTON,

BOOKSELLERS AND PUBLISHERS TO UNIVERSITY COLLEGE.

28 UPPER GOWER STREET.

M.DCCC.XXXIX.

FIRST NOTIONS

OF

LOGIC

(PREPARATORY TO THE STUDY OF GEOMETRY)

BY

AUGUSTUS DE MORGAN,

OF TRINITY COLLEGE, CAMBRIDGE,
PROFESSOR OF MATHEMATICS IN UNIVERSITY COLLEGE, LONDON.

The root of all the mischief in the sciences, is this ; that falsely magnifying and admiring the powers of the mind, we seek not its real helps. — BACON.

LONDON :

PRINTED FOR TAYLOR AND WALTON,
BOOKSELLERS AND PUBLISHERS TO UNIVERSITY COLLEGE.
28 UPPER GOWER STREET.

M.DCCC.XXXIX.

The Project Gutenberg eBook of First notions of logic (preparatory to the study of geometry)

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: First notions of logic (preparatory to the study of geometry)

Author: Augustus De Morgan

Release date: December 26, 2021 [eBook #67017]

Most recently updated: October 18, 2024

Language: English

Original publication: United Kingdom: Taylor and Walton, 1839

Other information and formats: www.gutenberg.org/ebooks/67017

Credits: Richard Tonsing and the Online Distributed Proofreading Team at <https://www.pgdp.net> (This file was produced from images generously made available by The Internet Archive)

*** START OF THE PROJECT GUTENBERG EBOOK FIRST NOTIONS
OF LOGIC (PREPARATORY TO THE STUDY OF GEOMETRY) ***

Transcriber's Note:

The cover image was created by the transcriber and is placed in the public domain.

FIRST NOTIONS
OF
LOGIC
(PREPARATORY TO THE STUDY OF GEOMETRY)

BY
AUGUSTUS DE MORGAN,
OF TRINITY COLLEGE, CAMBRIDGE,
PROFESSOR OF MATHEMATICS IN UNIVERSITY COLLEGE,
LONDON.

The root of all the mischief in the sciences, is this; that falsely magnifying and admiring the powers of the mind, we seek not its real helps.—BACON.

LONDON:
PRINTED FOR TAYLOR AND WALTON,
BOOKSELLERS AND PUBLISHERS TO UNIVERSITY COLLEGE.
28 UPPER GOWER STREET.

M.DCCC.XXXIX.

*✱ This Tract contains no more than the author has found, from experience, to be much wanted by students who are commencing with Euclid. It will ultimately form an Appendix to his Treatise on Arithmetic.

The author would not, by any means, in presenting the minimum necessary for a particular purpose, be held to imply that he has given enough of the subject for all the ends of education. He has long regretted the neglect of logic; a science, the study of which would shew many of its opponents that the light esteem in which they hold it arises from those habits of inference which thrive best in its absence. He strongly recommends any student to whom this tract may be the first introduction of the subject, to pursue it to a much greater extent.

University College, Jan, 8, 1839.

LONDON:—PRINTED BY JAMES MOYES,
Castle Street, Leicester Square.

FIRST NOTIONS OF LOGIC.

What we here mean by Logic is the examination of that part of reasoning which depends upon the manner in which inferences are formed, and the investigation of general maxims and rules for constructing arguments, so that the conclusion may contain no inaccuracy which was not previously asserted in the premises. It has nothing to do with the truth of the facts, opinions, or presumptions, from which an inference is derived; but simply takes care that the inference shall certainly be true, if the premises be true. Thus, when we say that all men will die, and that all men are rational beings, and thence infer that some rational beings will die, the *logical* truth of this sentence is the same whether it be true or false that men are mortal and rational. This logical truth depends upon the structure of the sentence, and not on the particular matters spoken of. Thus,

Instead of,	Write,
All men will die.	Every A is B.
All men are rational beings.	Every A is C.
Therefore some rational beings will die.	Therefore some Cs are Bs.

The second of these is the same proposition, logically considered, as the first; the consequence in both is virtually contained in, and rightly inferred from, the premises. Whether the premises be true or false, is not a question of logic, but of morals, philosophy, history, or any other knowledge to

which their subject-matter belongs: the question of logic is, does the conclusion certainly follow if the premises be true?

Every act of reasoning must mainly consist in comparing together different things, and either finding out, or recalling from previous knowledge, the points in which they resemble or differ from each other. That particular part of reasoning which is called *inference*, consists in the comparison of several and different things with one and the same other thing; and ascertaining the resemblances, or differences, of the several things, by means of the points in which they resemble, or differ from, the thing with which all are compared.

There must then be some propositions already obtained before any inference can be drawn. All propositions are either assertions or denials, and are thus divided into *affirmative* and *negative*. Thus, A is B, and A is not B, are the two forms to which all propositions may be reduced. These are, for our present purpose, the most simple forms; though it will frequently happen that much circumlocution is needed to reduce propositions to them. Thus, suppose the following assertion, ‘If he should come to-morrow, he will probably stay till Monday’; how is this to be reduced to the form A is B? There is evidently something spoken of, something said of it, and an affirmative connexion between them. Something, if it happen, that is, the happening of something, makes the happening of another something probable; or is one of the things which render the happening of the second thing probable.

A	is	B
---	----	---

The happening of his arrival to-morrow	is	an event from which it may be inferred as probable that he will stay till Monday.
---	----	--

The forms of language will allow the manner of asserting to be varied in a great number of ways; but the reduction to the preceding form is always possible. Thus, ‘so he said’ is an affirmation, reducible as follows:

What you have just said (or whatever else ‘so’ refers to)	is	the thing which he said.
--	----	-----------------------------

By changing ‘is’ into ‘is not,’ we make a negative proposition; but care must always be taken to ascertain whether a proposition which appears negative is really so. The principal danger is that of confounding a

proposition which is negative with another which is affirmative of something requiring a negative to describe it. Thus 'he resembles the man who was not in the room,' is affirmative, and must not be confounded with 'he does not resemble the man who was in the room.' Again, 'if he should come to-morrow, it is probable he will not stay till Monday,' does not mean the simple denial of the preceding proposition, but the affirmation of the directly opposite proposition. It is,

A	is	B
---	----	---

The happening of his arrival to-morrow,	is	an event from which it may be inferred to be <i>improbable</i> that he will stay till Monday,
--	----	---

whereas the following,

The happening of his arrival to-morrow,	is <i>not</i>	an event from which it may be inferred as probable that he will stay till Monday,
--	------------------	---

would be expressed thus: 'If he should come to-morrow, that is no reason why he should stay till Monday.'

Moreover, the negative words not, no, &c., have two kinds of meaning which must be carefully distinguished. Sometimes they deny, and nothing more: sometimes they are used to affirm the direct contrary. In cases which offer but two alternatives, one of which is necessary, these amount to the same thing, since the denial of one, and the affirmation of the other, are obviously equivalent propositions. In many idioms of conversation, the negative implies affirmation of the contrary in cases which offer not only alternatives, but degrees of alternatives. Thus, to the question, 'Is he tall?' the simple answer, 'No,' most frequently means that he is the contrary of tall, or considerably under the average. But it must be remembered, that, in all logical reasoning, the negation is simply negation, and nothing more, never implying affirmation of the contrary.

The common proposition that two negatives make an affirmative, is true only upon the supposition that there are but two possible things, one of which is denied. Grant that a man must be either able or unable to do a particular thing, and then *not unable* and able are the same things. But if we suppose various degrees of performance, and therefore degrees of ability, it

is false, in the common sense of the words, that two negatives make an affirmative. Thus, it would be erroneous to say, 'John is able to translate Virgil, and Thomas is not unable; therefore, what John can do Thomas can do,' for it is evident that the premises mean that John is so near to the best sort of translation that an affirmation of his ability may be made, while Thomas is considerably lower than John, but not so near to absolute deficiency that his ability may be altogether denied. It will generally be found that two negatives imply an affirmative of a weaker degree than the positive affirmation.

Each of the propositions, 'A is B,' and 'A is not B,' may be subdivided into two species: the *universal*, in which every possible case is included; and the *particular*, in which it is not meant to be asserted that the affirmation or negation is universal. The four species of propositions are then as follows, each being marked with the letter by which writers on logic have always distinguished it.

A <i>Universal Affirmative</i>	Every A is	B
E <i>Universal Negative</i>	No A is	B
I <i>Particular Affirmative</i>	Some A is	B
O <i>Particular Negative</i>	Some A is not	B

In common conversation the affirmation of a part is meant to imply the denial of the remainder. Thus, by 'some of the apples are ripe,' it is always intended to signify that some are not ripe. This is not the case in logical language, but every proposition is intended to make its amount of affirmation or denial, and no more. When we say, 'Some A is B,' or, more grammatically, 'Some As are Bs,' we do not mean to imply that some are not: this may or may not be. Again, the word some means, 'one or more, possibly all.' The following table will shew the bearing of each proposition on the rest.

Every A is B affirms and contains Some A is B and denies	No A is B Some A is not B
No A is B affirms and contains Some A is not B and denies	Every A is B Some A is B

<i>Some A is B</i> does not contradict	<i>Every A is B</i> <i>Some A is not B</i>	but denies <i>No A is B</i>
<i>Some A is not B</i> does not contradict	<i>No A is B</i> <i>Some A is B</i>	but denies <i>Every A is B</i>

Contradictory propositions are those in which one denies *any thing* that the other affirms; *contrary* propositions are those in which one denies *every thing* which the other affirms, or affirms *every thing* which the other denies. The following pair are contraries.

Every A is B and No A is B

and the following are contradictories,

Every A is B to Some A is not B
No A is B to Some A is B

A contrary, therefore, is a complete and total contradictory; and a little consideration will make it appear that the decisive distinction between contraries and contradictories lies in this, that contraries may both be false, but of contradictories, one must be true and the other false. We may say, 'Either P is true, or *something* in contradiction of it is true;' but we cannot say, 'Either P is true, or *every thing* in contradiction of it is true.' It is a very common mistake to imagine that the *denial* of a proposition gives a right to *affirm* the contrary; whereas it should be, that the *affirmation* of a proposition gives a right to *deny* the contrary. Thus, if we deny that Every A is B, we do not affirm that No A is B, but only that Some A is not B; while, if we affirm that Every A is B, we deny No A is B, and also Some A is not B.

But, as to contradictories, affirmation of one is a denial of the other, and denial of one is affirmation of the other. Thus, either Every A is B, or Some A is not B: affirmation of either is denial of the other, and *vice versá*.

Let the student now endeavour to satisfy himself of the following. Taking the four preceding propositions, A, E, I, O, let the simple letter signify the

affirmation, the same letter in parentheses the denial, and the absence of the letter, that there is neither affirmation nor denial.

From A	follow	(E), I, (O)	From (A)	follow	O
From E		(A), (I), O	From (E)		I
From I		(E)	From (I)		(A), E, O
From O		(A)	From (O)		A, (E), I

These may be thus summed up: The affirmation of a universal proposition, and the denial of a particular one, enable us to affirm or deny all the other three; but the denial of a universal proposition, and the affirmation of a particular one, leave us unable to affirm or deny two of the others.

In such propositions as ‘Every A is B,’ ‘Some A is not B,’ &c., A is called the *subject*, and B the *predicate*, while the verb ‘is’ or ‘is not,’ is called the *copula*. It is obvious that the words of the proposition point out whether the subject is spoken of universally or partially, but not so of the predicate, which it is therefore important to examine. Logical writers generally give the name of *distributed* subjects or predicates to those which are spoken of universally; but as this word is rather technical, I shall say that a subject or predicate enters wholly or partially, according as it is universally or particularly spoken of.

1. In A, or ‘Every A is B,’ the subject enters wholly, but the predicate only partially. For it obviously says, ‘Among the Bs are all the As,’ ‘Every A is part of the collection of Bs, so that all the As make a part of the Bs, the whole it *may* be.’ Thus, ‘Every horse is an animal,’ does not speak of all animals, but states that all the horses make up a portion of the animals.

2. In E, or ‘No A is B,’ both subject and predicate enter wholly. ‘No A whatsoever is any one out of all the Bs;’ ‘search the whole collection of Bs, and *every* B shall be found to be something which is not A.’

3. In I, or ‘Some A is B,’ both subject and predicate enter partially. ‘Some of the As are found among the Bs, or make up a part (the whole possibly, but not known from the preceding) of the Bs.’

4. In O, or ‘Some A is not B,’ the subject enters partially, and the predicate wholly. ‘Some As are none of them any whatsoever of the Bs; every B will be found to be no one out of a certain portion of the As.’

It appears then that,

In affirmatives, the predicate enters partially.

In negatives, the predicate enters wholly.

In contradictory propositions, both subject and predicate enter differently in the two.

The *converse* of a proposition is that which is made by interchanging the subject and predicate, as follows:

The proposition.	Its converse.
A Every A is B	Every B is A
E No A is B	No B is A
I Some A is B	Some B is A
O Some A is not B	Some B is not A

Now, it is a fundamental and self-evident proposition, that no consequence must be allowed to assert more widely than its premises; so that, for instance, an assertion which is only of some Bs can never lead to a result which is true of all Bs. But if a proposition assert agreement or disagreement, any other proposition which asserts the same, to the same extent and no further, must be a legitimate consequence; or, if you please, must amount to the whole, or part, of the original assertion in another form. Thus, the converse of A is not true: for, in 'Every A is B,' the predicate enters partially; while in 'Every B is A,' the subject enters wholly. 'All the As make up a part of the Bs, then a part of the Bs are among the As, or some B is A.' Hence, the only *legitimate* converse of 'Every A is B' is, 'Some B is A.' But in 'No A is B,' both subject and predicate enter wholly, and 'No B is A' is, in fact, the same proposition as 'No A is B.' And 'Some A is B' is also the same as its converse 'Some B is A;' here both terms enter partially. But 'Some A is not B' admits of no converse whatever; it is perfectly consistent with all assertions upon B and A in which B is the subject. Thus neither of the four following lines is inconsistent with itself.

Some A is not B and Every B is A
 Some A is not B and No B is A
 Some A is not B and Some B is A
 Some A is not B and Some B is not A.

We find then, including converses, which are not identical with their direct propositions, six different ways of asserting or denying, with respect to agreement or non-agreement, total or partial, between A and, say X: these we write down, designating the additional assertions by U and Y.

A Every A is X U Every X is A	E	Identical.		I	Identical.		O Some A is not X Y Some X is not A
		No A is X No X is A			Some A is X Some X is A		

We shall now repeat and extend the table of page 8 (A), &c., meaning, as before, the denial of A, &c.

From A or (O)	follow	A,	(E),	I	(O)		
From E or (I)		(A),	E,	(I),	O,	(U),	Y
From I or (E)			(E)	I			
From O or (A)		(A),			O		
From U or (Y)			(E)	I,		U	(Y)
From Y or (U)						(U)	Y

Having thus discussed the principal points connected with the simple assertion, we pass to the manner of making two assertions give a third. Every instance of this is called a syllogism, the two assertions which form the basis of the third are called premises, and the third itself the conclusion.

If two things both agree with a third in any particular, they agree with each other in the same; as, if A be of the same colour as X, and B of the same colour as X, then A is of the same colour as B. Again, if A differ from X in any particular in which B agrees with X, then A and B differ in that particular. If A be not of the same colour as X, and B be of the same colour as X, then A is not of the colour of B. But if A and B both differ from X in

any particular, nothing can be inferred; they may either differ in the same way and to the same extent, or not. Thus, if A and B be both of different colours from X, it neither follows that they agree, nor differ, in their own colours.

The paragraph preceding contains the essential parts of all inference, which consists in comparing two things with a third, and finding from their agreement or difference with that third, their agreement or difference with one another. Thus, Every A is X, every B is X, allows us to infer that A and B have all those qualities in common which are necessary to X. Again, from Every A is X, and 'No B is X,' we infer that A and B differ from one another in all particulars which are essential to X. The preceding forms, however, though they represent common reasoning better than the ordinary syllogism, to which we are now coming, do not constitute the ultimate forms of inference. Simple *identity* or *non-identity* is the ultimate state to which every assertion may be reduced; and we shall, therefore, first ask, from what identities, &c., can other identities, &c., be produced? Again, since we name objects in species, each species consisting of a number of individuals, and since our assertion may include all or only part of a species, it is further necessary to ask, in every instance, to what extent the conclusion drawn is true, whether of all, or only of part?

Let us take the simple assertion, 'Every living man respire;' or, every living man is one of the things (however varied they may be) which respire. If we were to inclose all living men in a large triangle, and all respiring objects in a large circle, the preceding assertion, if true, would require that the whole of the triangle should be contained in the circle. And in the same way we may reduce any assertion to the expression of a coincidence, total or partial, between two figures. Thus, a point in a circle may represent an individual of one species, and a point in a triangle an individual of another species: and we may express that the whole of one species is asserted to be contained or not contained in the other by such forms as, 'All the $\triangle$ is in the $\circ$ '; 'None of the $\triangle$ is in the $\circ$ '.

Any two assertions about A and B, each expressing agreement or disagreement, total or partial, with or from X, and leading to a conclusion with respect to A or B, is called a syllogism, of which X is called the *middle term*. The plainest syllogism is the following:—

Every A is X		All the $\triangle$ is in the $\circ$
Every X is B		All the $\circ$ is in the $\square$
Therefore Every A is B	Therefore	All the $\triangle$ is in the $\square$

In order to find all the possible forms of syllogism, we must make a table of all the elements of which they can consist; namely—

A and X		B and X	
Every A is X	A	Every B is X	
No A is X	E	No B is X	
Some A is X	I	Some B is X	
Some A is not X	O	Some B is not X	
Every X is A	U	Every X is B	
Some X is not A	Y	Some X is not B	

Or their synonymes,

$\triangle$ and $\circ$		$\square$ and $\circ$	
All the $\triangle$ is in the $\circ$	A	All the $\square$ is in the $\circ$	
None of the $\triangle$ is in the $\circ$	E	None of the $\square$ is in the $\circ$	
Some of the $\triangle$ is in the $\circ$	I	Some of the $\square$ is in the $\circ$	
Some of the $\triangle$ is not in the $\circ$	O	Some of the $\square$ is not in the $\circ$	
All the $\circ$ is in the $\triangle$	U	All the $\circ$ is in the $\square$	
Some of the $\circ$ is not in the $\triangle$	Y	Some of the $\circ$ is not in the $\square$	

Now, taking any one of the six relations between A and X, and combining it with either of those between B and X, we have six pairs of premises, and the same number repeated for every different relation of A and X. We have then thirty-six forms to consider: but, thirty of these (namely, all but (A, A) (E, E), &c.) are half of them repetitions of the other half. Thus, 'Every A is X, no B is X,' and 'Every B is X, no A is X,' are of the same form, and only differ by changing A into B and B into A. There are then only 15 + 6, or 21 distinct forms, some of which give a necessary conclusion, while others do

not. We shall select the former of these, classifying them by their conclusions; that is, according as the inference is of the form A, E, I, or O.

I. In what manner can a universal affirmative conclusion be drawn; namely, that one figure is entirely contained in the other? This we can only assert when we know that one figure is entirely contained in the circle, which itself is entirely contained in the other figure. Thus,

Every A is X		All the $\triangle$ is in the $\circ$	A
Every X is B		All the $\circ$ is in the $\square$	A
$\therefore$ Every A is B		$\therefore$ All the $\triangle$ is in the $\square$	A

is the only way in which a universal affirmative conclusion can be drawn.

II. In what manner can a universal negative conclusion be drawn; namely, that one figure is entirely exterior to the other? Only when we are able to assert that one figure is entirely within, and the other entirely without, the circle. Thus,

Every A is X		All the $\triangle$ is in the $\circ$	A
No B is X		None of the $\square$ is in the $\circ$	E
$\therefore$ No A is B		$\therefore$ None of the $\triangle$ is in the $\square$	E

is the only way in which a universal negative conclusion can be drawn.

III. In what manner can a particular affirmative conclusion be drawn; namely, that part or all of one figure is contained in the other? Only when we are able to assert that the whole circle is part of one of the figures, and that the whole, or part of the circle, is part of the other figure. We have then two forms.

Every X is A		All the $\circ$ is in the $\triangle$	A
Every X is B		All the $\circ$ is in the $\square$	A
$\therefore$ Some A is B		$\therefore$ Some of the $\triangle$ is in the $\square$	I
Every X is A		All the $\circ$ is in the $\triangle$	A
Some X is B		Some of the $\circ$ is in the $\square$	I
Some A is B		Some of the $\triangle$ is in the $\square$	I

The second of these contains all that is strictly necessary to the conclusion, and the first may be omitted. That which follows when an assertion can be made as to some, must follow when the same assertion can be made of all.

IV. How can a particular negative proposition be inferred; namely, that part, or all of one figure, is not contained in the other? It would seem at first sight, whenever we are able to assert that part or all of one figure is in the circle, and that part or all of the other figure is not. The weakest syllogism from which such an inference can be drawn would then seem to be as follows.

Some A is X	Some of the $\triangle$ is in the $\circ$
Some B is not X	Some of the $\square$ is not in the $\circ$
$\therefore$ Some B is not A	$\therefore$ Some of the $\triangle$ is not in the $\square$

But here it will appear, on a little consideration, that the conclusion is only thus far true; that those As which are Xs cannot be *those* Bs which are not Xs; but they may be *other* Bs, about which nothing is asserted when we say that *some* Bs are not Xs. And further consideration will make it evident, that a conclusion of this form can only be arrived at when one of the figures is entirely within the circle, and the whole or part of the other without; or else when the whole of one of the figures is without the circle, and the whole or part of the other within; or lastly, when the circle lies entirely within one of the figures, and not entirely within the other. That is, the following are the distinct forms which allow of a particular negative conclusion, in which it should be remembered that a particular proposition in the premises may always be changed into a universal one, without affecting the conclusion. For that which necessarily follows from “some,” follows from “all.”

Every A is X	All the $\triangle$ is in the $\circ$	A
Some B is not X	Some of the $\square$ is not in the $\circ$	O
$\therefore$ Some B is not A	Some of the $\square$ is not in the $\triangle$	O

No A Is X	None of the $\triangle$ is in the $\circ$	E
Some B is X	Some of the $\square$ is in the $\circ$	I
$\therefore$ Some B is not A	Some of the $\square$ is not in the $\triangle$	O

Every X is A	All the $\circ$ is in the $\triangle$	A
Some X is not B	Some of the $\circ$ is not in the $\square$	O
$\therefore$ Some A is not B	Some of the $\triangle$ is not in the $\square$	O

It appears, then, that there are but six distinct syllogisms. All others are made from them by strengthening one of the premises, or converting one or both of the premises, where such conversion is allowable; or else by first making the conversion, and then strengthening one of the premises. And the following arrangement will shew that two of them are universal, three of the others being derived from them by weakening one of the premises in a manner which does not destroy, but only weakens, the conclusion.

1. Every A is X Every X is B		3. Every A is X No B is X	
-----		-----	
Every A is B		No A Is B	
2. Some A is X Every X is B	4. Some A is X No B is X	5. Every A is X Some B is not X	6. Every X is A Some X is not B
-----	-----	-----	-----
Some A is B	Some A is not B	Some B is not A	Some A is not B

We may see how it arises that one of the partial syllogisms is not immediately derived, like the others, from a universal one. In the preceding, AEE may be considered as derived from AAA, by changing the term in which X enters universally into its contrary. If this be done with the other term instead, we have

No A is		from which universal premises we cannot deduce a universal conclusion, but only Some B is not A.
X		
Every X is B		

If we weaken one and the other of these premises, as they stand, we obtain

Some A is not X		No A is X
Every X is B	and	Some X is B
No conclusion		Some B is not A

equivalent to the fourth of the preceding: but if we convert the first premiss, and proceed in the same manner,

	No X is A		Some X is not A
From	Every X is B	we obtain	Every X is B
	Some B is not A		Some B is not A

which is legitimate, and is the same as the last of the preceding list, with A and B interchanged.

Before proceeding to shew that all the usual forms are contained in the preceding, let the reader remark the following rules, which may be proved either by collecting them from the preceding cases, or by independent reasoning.

1. The middle term must enter universally into one or the other premiss. If it were not so, the one premiss might speak of one part of the middle term, and the other of the other; so that there would, in fact, be no middle term. Thus, 'Every A is X, Every B is X,' gives no conclusion: it may be thus stated;

All the As make up *a part* of the Xs
 All the Bs make up *a part* of the Xs

And, before we can know that there is any common term of comparison at all, we must have some means of shewing that the two parts are the same; or the preceding premises by themselves are inconclusive.

2. No term must enter the conclusion more generally than it is found in the premises; thus, if A be spoken of partially in the premises, it must enter partially into the conclusion. This is obvious, since the conclusion must assert no more than the premises imply.

3. From premises both negative no conclusion can be drawn. For it is obvious, that the mere assertion of disagreement between each of two things and a third, can be no reason for inferring either agreement or disagreement between these two things. It will not be difficult to reduce any case which falls under this rule to a breach of the first rule: thus, No A is X, No B is X, gives

Every A is (something which is not X)
Every B is (something which is not X)

in which the middle term is not spoken of universally in either. Again, 'No X is A, Some X is not B,' may be converted into

Every A is (a thing which is not X)
Some (thing which is not B) is X

in which there is no middle term.

4. From premises both particular no conclusion can be drawn. This is sufficiently obvious when the first or second rule is broken, as in 'Some A is X, Some B is X.' But it is not immediately obvious when the middle term enters one of the premises universally. The following reasoning will serve for exercise in the preceding results. Since both premises are particular in form, the middle term can only enter one of them universally by being the predicate of a negative proposition; consequently (Rule 3) the other premiss must be affirmative, and, being particular, neither of its terms is universal. Consequently both the terms as to which the conclusion is to be drawn enter partially, and the conclusion (Rule 2) can only be a particular *affirmative* proposition. But if one of the premises be negative, the conclusion must be *negative* (as we shall immediately see). This contradiction shews that the supposition of particular premises producing a legitimate result is inadmissible.

5. If one premiss be negative, the conclusion, if any, must be negative. If one term agree with a second and disagree with a third, no agreement can be inferred between the second and third.

6. If one premiss be particular, the conclusion must be particular. This is not very obvious, since the middle term may be universally spoken of in a particular proposition, as in Some B is not X. But this requires one negative proposition, whence (Rule 3) the other must be affirmative. Again, since the conclusion must be negative (Rule 5) its predicate is spoken of universally, and, therefore, must enter universally; the other term A must enter, then, in a universal affirmative proposition, which is against the supposition.

In the preceding set of syllogisms we observe one form only which produces A, or E, or I, but three which produce O.

Let an assertion be said to be weakened when it is reduced from universal to particular, and strengthened in the contrary case. Thus, 'Every A is B' is called stronger than 'Some A is B.'

Every form of syllogism which can give a legitimate result is either one of the preceding six, or another formed from one of the six, either by changing one of the assertions into its converse, if that be allowable, or by strengthening one of the premises without altering the conclusion, or both. Thus,

Some A is X		Some X is A
Every X is B	may be written	Every X is B
What follows will still follow from		<i>Every X is A</i>
		Every X is B

for all which is true when 'Some X is A,' is not less true when 'Every X is A.'

It would be possible also to form a legitimate syllogism by weakening the conclusion, when it is universal, since that which is true of all is true of some. Thus, 'Every A is X, Every X is B,' which yields 'Every A is B,' also yields 'Some A is B.' But writers on logic have always considered these syllogisms as useless, conceiving it better to draw from any premises their strongest conclusion. In this they were undoubtedly right; and the only question is, whether it would not have been advisable to make the premises as weak as possible, and not to admit any syllogisms in which more

appeared than was absolutely necessary to the conclusion. If such had been the practice, then

Every X is A, Every X is B, therefore Some A is B

would have been considered as formed by a spurious and unnecessary excess of assertion. The minimum of assertion would be contained in either of the following,

Every X is A, Some X is B, therefore Some A is B

Some X is A, Every X is B, therefore Some A is B

In this tract, syllogisms have been divided into two classes: first, those which prove a universal conclusion; secondly, those which prove a partial conclusion, and which are (all but one) derived from the first by weakening one of the premises, in such manner as to produce a legitimate but weakened conclusion. Those of the first class are placed in the first column, and the other in the second.

Universal.		Particular.	
A	Every A is X	Some A is X	I
A	Every X is B	Every X is B	A
A	Every A is B	Some A is B	I
		Some A is X	I
		No X is B	E
A	Every A is X	Some A is not B	O
E	No X is B	Every A is X	A
E	No A is B	Some B is not X	O
		Some B is not A	O
		Every X is A	A
		Some X is not B	O
		Some A is not B	O

In all works on logic, it is customary to write that premiss first which contains the predicate of the conclusion. Thus,

Every X is B		Every A is X
Every A is X	would be written, and not	Every X is B
Every A is B		Every A is B

The premises thus arranged are called major and minor; the predicate of the conclusion being called the major term, and its subject the minor. Again, in the preceding case we see the various subjects coming in the order X, B; A, X; A, B: and the number of different orders which can appear is four, namely—

XB	BX	XB	BX
AX	AX	XA	XA
AB	AB	AB	AB

which are called the four figures, and every kind of syllogism in each figure is called a mood. I now put down the various moods of each figure, the letters of which will be a guide to find out those of the preceding list from which they are derived. Co means that a premiss of the preceding list has been converted; + that it has been strengthened; Co +, that both changes have taken place. Thus,

A	Every X is B		A	Every X is B
I	Some A is X	becomes	A	Every X is A: (Co +)
I	Some A is B		I	Some A is B

And Co + abbreviates the following: If some A be X, then some X is A (Co); and all that is true when Some X is A, is true when Every X is A (+); therefore the second is legitimate, if the first be so.

First Figure.

A Every X is
B

A Every A is
X

A Every A is B

E No X is B

A Every A is
X

E No A is B

A Every X is B

I Some A is X

I Some A is B

E No X is B

I Some A is X

O Some A is not B

Second Figure.

E No B is X (Co)

A Every A is
X

E No A is B

A Every B is
X

E No A is X (Co)

E No A is B

E No B is X (Co)

I Some A is X

O Some A is not B

A Every B is X

O Some A is not X

O Some A is not B

Third Figure.

A Every X is
B

A Every X is (Co+)
A

I Some A is B

I Some X is B (Co)
A Every X is
A

I Some A is B

A Every X is
B

I Some X is A (Co)

I Some A is B

E No X is B

A Every X is A
(Co+)

O Some A is not B

O Some X is not B
A Every X is A

O Some A is not B

E No X is B

I Some X is A (Co)

O Some A is not B

Fourth Figure.

A Every B is (+)
X

A Every X is
A

I Some A is B

I Some B is X

A Every X is A

I Some B is A

A Every B is
X

E No X is A

E No A is B

E No B is X (Co)

A Every X is A
(Co+)

O Some A is not B

E No B is X (Co)

I Some X is A
(Co)

O Some A is not B

The above is the ancient method of dividing syllogisms; but, for the present purpose, it will be sufficient to consider the six from which the rest can be obtained. And since some of the six have A in the predicate of the conclusion, and not B, we shall join to them the six other syllogisms which are found by transposing B and A. The complete list, therefore, of syllogisms with the weakest premises and the strongest conclusions, in which a comparison of A and B is obtained by comparison of both with X, is as follows:

Every A is X	Every B is X	Some A is X	Some B is X
Every X is B	Every X is A	No X is B	No X is A
Every A is B	Every B is A	Some A is not B	Some B is not A
Every A is X	Every B is X	Every A is X	Every B is X
No X is B	No X is A	Some B is not X	Some A is not X
No A is B	No B is A	Some B is not A	Some A is not B
Some A is X	Some B is X	Every X is A	Every X is B
Every X is B	Every X is A	Some X is not B	Some X is not A
Some A is B	Some B is A	Some A is not B	Some B is not A

In the list of page [19](#), there was nothing but recapitulation of forms, each form admitting a variation by interchanging A and B. This interchange having been made, and the results collected as above, if we take every case in which B is the predicate, or can be made the predicate by allowable conversion, we have a collection of all possible *weakest* forms in which the result is one of the four ‘Every A is B,’ ‘No A is B,’ ‘Some A is B,’ ‘Some A is not B’; as follows. The premises are written in what appeared the most natural order, without distinction of major or minor; and the letters prefixed are according to the forms of the several premises, as in page [10](#).

A Every A is
X

U Every X is
B

A Every A is
B

I Some A is
X

U Every X
is B

I Some A is
B

A Every A
is X

E No B is X

E No A is B

I Some B
is X

U Every X
is A

I Some A is
B

A Every B
is X

E No A is X

E No A is B

I Some A is
X

E No B is X

O Some A is
not B

A Every B is
X

O Some A is
not X

O Some A is
not B

U Every X is
A

Y Some X is
not B

O Some A is
not B

Every assertion which can be made upon two things by comparison with any third, that is, every simple inference, can be reduced to one of the

preceding forms. Generally speaking, one of the premises is omitted, as obvious from the conclusion; that is, one premiss being named and the conclusion, that premiss is implied which is necessary to make the conclusion good. Thus, if I say, “That race must have possessed some of the arts of life, for they came from Asia,” it is obviously meant to be asserted, that all races coming from Asia must have possessed some of the arts of life. The preceding is then a syllogism, as follows:

‘That race’ is ‘a race of Asiatic origin:’

Every ‘race of Asiatic origin’ is ‘a race which must have possessed some of the arts of life:’

Therefore, That race is a race which must have possessed some of the arts of life.

A person who makes the preceding assertion either means to imply, antecedently to the conclusion, that all Asiatic races must have possessed arts, or he talks nonsense if he asserts the conclusion positively. ‘A must be B, for it is X,’ can only be true when ‘Every X is B.’ This latter proposition may be called the suppressed premiss; and it is in such suppressed propositions that the greatest danger of error lies. It is also in such propositions that men convey opinions which they would not willingly express. Thus, the honest witness who said, ‘I always thought him a respectable man—he kept his gig,’ would probably not have admitted in direct terms, ‘Every man who keeps a gig must be respectable.’

I shall now give a few detached illustrations of what precedes.

“His imbecility of character might have been inferred from his proneness to favourites; for all weak princes have this failing.” The preceding would stand very well in a history, and many would pass it over as containing very good inference. Written, however, in the form of a syllogism, it is,

	All weak princes	are prone to favourites
	He	was prone to favourites
Therefore	He	was a weak prince

which is palpably wrong. (Rule 1.) The writer of such a sentence as the preceding might have meant to say, ‘for all who have this failing are weak princes;’ in which case he would have inferred rightly. Every one should be aware that there is much false inference arising out of badness of style, which is just as injurious to the habits of the untrained reader as if the errors were mistakes of logic in the mind of the writer.

‘A is less than B; B is less than C: therefore A is less than C.’ This, at first sight, appears to be a syllogism; but, on reducing it to the usual form, we find it to be,

	A is (a magnitude less than B)
	B is (a magnitude less than C)
Therefore	A is (a magnitude less than C)

which is not a syllogism, since there is no middle term. Evident as the preceding is, the following additional proposition must be formed before it can be made explicitly logical. ‘If B be a magnitude less than C, then every magnitude less than B is also less than C.’ There is, then, before the preceding can be reduced to a syllogistic form, the necessity of a deduction from the second premiss, and the substitution of the result instead of that premiss. Thus,

	A is less than	
	B	
Less	B is less than	following from B is less than
than	C:	C.

Therefore	A is less than	
	C	

But, if the additional argument be examined—namely, if B be less than C, then that which is less than B is less than C—it will be found to require precisely the same considerations repeated; for the original inference was nothing more. In fact, it may easily be seen as follows, that the proposition

before us involves more than any simple syllogism can express. When we say that A is less than B, we say that if A were applied to B, every part of A would match a part of B, and there would be parts of B remaining over. But when we say, 'Every A is B,' meaning the premiss of a common syllogism, we say that every instance of A is an instance of B, without saying any thing as to whether there are or are not instances of B still left, after those which are also A are taken away. If, then, we wish to write an ordinary syllogism in a manner which shall correspond with 'A is less than B, B is less than C, therefore A is less than C,' we must introduce a more definite amount of assertion than was made in the preceding forms. Thus,

Every A is B, and there are Bs which are not As
 Every B is C, and there are Cs which are not Bs

Therefore Every A is C, and there are Cs which are not As

Or thus:

The Bs contain all the As, and more
 The Cs contain all the Bs, and more

The Cs contain all the As, and more

The most technical form, however, is,

From Every A is B; [Some B is not A]
 Every B is C; [Some C is not B]
 Follows Every A is C; [Some C is not A]

This sort of argument is called *à fortiori* argument, because the premises are more than sufficient to prove the conclusion, and the extent of the conclusion is thereby greater than its mere form would indicate. Thus, 'A is less than B, B is less than C, therefore, *à fortiori*, A is less than C,' means

that the extent to which A is less than C must be greater than that to which A is less than B, or B than C. In the syllogism last written, either of the bracketed premises might be struck out without destroying the conclusion; which last would, however, be weakened. As it stands, then, the part of the conclusion, 'Some C is not A,' follows it *à fortiori*.

The argument *à fortiori*, may then be defined as a universally affirmative syllogism, in which both of the premises are shewn to be less than the whole truth, or greater. Thus, in 'Every A is X, Every X is B, therefore Every A is B,' we do not certainly imply that there are more Xs than As, or more Bs than Xs, so that we do not know that there are more Bs than As. But if we are at liberty to state the syllogism as follows,

All the As make up part (and part only) of the Xs
Every X is B;

then we are certain that

All the As make up part (and part only) of the Bs.

But if we are at liberty further to say that

All the As make up part (and part only) of the Xs
All the Xs make up part (and part only) of the Bs

then we conclude that

All the As make up *part of part* (only) of the Bs

and the words in Italics mark that quality of the conclusion from which the argument is called *à fortiori*.

Most syllogisms which give an affirmative conclusion are generally meant to imply *à fortiori* arguments, except only in mathematics. It is seldom, except in the exact sciences, that we meet with a proposition, 'Every A is B,' which we cannot immediately couple with 'Some Bs are not As.'

When an argument is completely established, with the exception of one assertion only, so that the inference may be drawn as soon as that one assertion is established, the result is stated in a form which bears the name of an *hypothetical* syllogism. The word hypothesis means nothing but supposition; and the species of syllogism just mentioned first lays down the

assertion that a consequence will be true if a certain condition be fulfilled, and then either asserts the fulfilment of the condition, and thence the consequence, or else denies the consequence, and thence denies the fulfilment of the condition. Thus, if we know that

When A is B, it follows that P is Q;

then, as soon as we can ascertain that A is B, we can conclude that P is Q; or, if we can shew that P is not Q, we know that A is not B. But if we find that A is not B, we can infer nothing; for the preceding does not assert that P is Q *only* when A is B. And if we find out that P is Q, we can infer nothing. This conditional syllogism may be converted into an ordinary syllogism, as follows. Let K be any 'case in which A is B,' and Z a 'case in which P is Q'; then the preceding assertion amounts to 'Every K is Z.' Let L be a particular instance, the A of which may or may not be B. If A be B in the instance under discussion, or if A be not B, we have, in the one case and the other,

	Every	K is Z	Every	K is Z
		L is a K		L is not a K
		—————		—————
Therefore		L is a Z		No conclusion

Similarly, according as a particular case (M) is or is not Z, we have

Every	K is Z	Every	K is Z
	M is a Z		M is not a Z
	—————		—————
	No conclusion		M is not a K

That is to say: The assertion of an hypothesis is the assertion of its necessary consequence, and the denial of the necessary consequence is the denial of the hypothesis; but the assertion of the necessary consequence gives no right to assert the hypothesis, nor does the denial of the hypothesis give any right to deny the truth of that which would (were the hypothesis true) be its necessary consequence.

Demonstration is of two kinds: which arises from this, that every proposition has a contradictory; and of these two, one must be true and the other must be false. We may then either prove a proposition to be true, or its contradictory to be false. 'It is true that Every A is B,' and, 'it is false that there are some As which are not Bs,' are the same proposition; and the proof of either is called the indirect proof of the other.

But how is any proposition to be proved false, except by proving a contradiction to be true? By proving a necessary consequence of the proposition to be false. But this is not a complete answer, since it involves the necessity of doing the same thing; or, so far as this answer goes, one proposition cannot be proved false unless by proving another to be false. But it may happen, that a necessary consequence can be obtained which is obviously and self-evidently false, in which case no further proof of the falsehood of the hypothesis is necessary. Thus the proof which Euclid gives that all equiangular triangles are equilateral is of the following structure, logically considered.

(1.) If there be an equiangular triangle not equilateral, it follows that a whole can be found which is not greater than its part.^[1]

^{1.} This is the proposition in proof of which nearly the whole of the demonstration of Euclid is spent.

(2.) It is false that there can be any whole which is not greater than its part (self evident).

(3.) Therefore it is false that there is any equiangular triangle which is not equilateral; or all equiangular triangles are equilateral.

When a proposition is established by proving the truth of the matters it contains, the demonstration is called *direct*; when by proving the falsehood of every contradictory proposition, it is called *indirect*. The latter species of demonstration is as logical as the former, but not of so simple a kind; whence it is desirable to use the former whenever it can be obtained.

The use of indirect demonstration in the Elements of Euclid is almost entirely confined to those propositions in which the converses of simple propositions are proved. It frequently happens that an established assertion of the form

Every A is B (1)

may be easily made the means of deducing,

Every (thing not A) is not B (2)

which last gives

Every B is A (3)

The conversion of the second proposition into the third is usually made by an indirect demonstration, in the following manner. If possible, let there be one B which is not A, (2) being true. Then there is one thing which is not A and is B; but every thing not A is not B; therefore there is one thing which is B and is not B: which is absurd. It is then absurd that there should be one single B which is not A; or, Every B is A.

The following proposition contains a method which is of frequent use.

HYPOTHESIS.—Let there be any number of propositions or assertions,—three for instance, A, B, and C,—of which it is the property that one or the other must be true, *and one only*. Let there be three other propositions, P, Q, and R, of which it is also the property that one, and one only, must be true. Let it also be a connexion of those assertions, that

When A is true, P is true

When B is true, Q is true

When C is true, R is true

CONSEQUENCE: then it follows that

When P is true, A is true

When Q is true, B is true

When R is true, C is true

For, when P is true, then Q and R must be false; consequently, neither B nor C can be true, for then Q or R would be true. But either A, B, or C must be

true, therefore A must be true; or, when P is true, A is true. In a similar way the remaining assertions may be proved.

Case 1. If	When P is Q,	A is B
	When P is not Q,	A is not B
It follows that	When A is B,	P is Q
	When A is not B,	P is not Q
Case 2. If	When A is greater than B,	P is greater than Q
	When A is equal to B,	P is equal to Q
	When A is less than B,	P is less than Q
It follows that	When P is greater than Q,	A is greater than B
	When P is equal to Q,	A is equal to B
	When P is less than Q,	A is less than B

We have hitherto supposed that the premises are actually true; and, in such a case, the logical conclusion is as certain as the premises. It remains to say a few words upon the case in which the premises are probably, but not certainly, true.

The probability of an event being about to happen, and that of an argument being true, may be so connected that the usual method of measuring the first may be made to give an easy method of expressing the second. Suppose an urn, or lottery, with a large number of balls, black or white; then, if there be twelve white balls to one black, we say it is twelve to one that a white ball will be drawn, or that a white ball is twelve times as probable as a black one. A certain assertion may be in the same condition as to the force of probability with which it strikes the mind: that is, the questions

Is the assertion true?

Will a white ball be drawn?

may be such that the answer, 'most probably,' expresses the same degree of likelihood in both cases.

We have before explained that logic has nothing to do with the truth or falsehood of assertions, but only professes, supposing them true, to collect

and classify the legitimate methods of drawing inferences. Similarly, in this part of the subject, we do not trouble ourselves with the question, How are we to find the probability due to premises? but we ask: Supposing (happen how it may) that we *have* found the probability of the premises, required the probability of the conclusion. When the odds in favour of a conclusion are, say 6 to 1, there are, out of every 7 possible chances, 6 in favour of the conclusion, and 1 against it. Hence $\frac{6}{7}$ and $\frac{1}{7}$ will represent the proportions, for and against, of all the possible cases which exist.

Thus we have the succession of such results as in the following table:—

Odds in favour of an event	Probability for	Probability against
1 to 1	$\frac{1}{2}$	$\frac{1}{2}$
2 to 1	$\frac{2}{3}$	$\frac{1}{3}$
3 to 1	$\frac{3}{4}$	$\frac{1}{4}$
3 to 2	$\frac{3}{5}$	$\frac{2}{5}$
4 to 1	$\frac{4}{5}$	$\frac{1}{5}$
4 to 3	$\frac{4}{7}$	$\frac{3}{7}$
5 to 1	$\frac{5}{6}$	$\frac{1}{6}$
&c.	&c.	&c.

Let the probability of a conclusion, as derived from the premises (that is on the supposition that it was never imagined to be possible till the argument was heard), be called the *intrinsic probability* of the argument. This is found by multiplying together the probabilities of all the assertions which are necessary to the argument. Thus, suppose that a conclusion was held to be impossible until an argument of a single syllogism was produced, the premises of which have severally five to one and eight to one in their favour. Then $\frac{5}{6} \times \frac{8}{9}$, or $\frac{40}{54}$, is the intrinsic probability of the argument, and the odds in its favour are 40 to 14, or 20 to 7.

But this intrinsic probability is not always that of the conclusion; the latter, of course, depending in some degree on the likelihood which the conclusion was supposed to have before the argument was produced. A syllogism of 20 to 7 in its favour, advanced in favour of a conclusion which

was beforehand as likely as not, produces a much more probable result than if the conclusion had been thought absolutely false until the argument produced a certain belief in the possibility of its being true. The change made in the probability of a conclusion by the introduction of an argument (or of a new argument, if some have already preceded) is found by the following rule.

From the sum of the existing probability of the conclusion and the intrinsic probability of the new argument, take their product; the remainder is the probability of the conclusion, as reinforced by the argument. Thus, $a + b - ab$ is the probability of the truth of a conclusion after the introduction of an argument of the intrinsic probability b , the previous probability of the said conclusion having been a .

Thus, a conclusion which has at present the chance $\frac{2}{3}$ in its favour, when reinforced by an argument whose intrinsic probability is $\frac{3}{4}$, acquires the probability $\frac{2}{3} + \frac{3}{4} - \frac{2}{3} \times \frac{3}{4}$ or, $\frac{2}{3} + \frac{3}{4} - \frac{1}{2}$, or $\frac{11}{12}$; or, having 2 to 1 in its favour before, it has 11 to 1 in its favour after, the argument.

When the conclusion was neither likely nor unlikely beforehand (or had the probability $\frac{1}{2}$), the shortest way of applying the preceding rule (in which $a + b - ab$ becomes $\frac{1}{2} + \frac{1}{2}b$) is to divide the sum of the numerator and denominator of the intrinsic probability of the argument by twice the denominator. Thus, an argument of which the intrinsic probability is $\frac{3}{4}$, gives to a conclusion on which no bias previously existed, the probability $\frac{7}{8}$ or $\frac{3+4}{2 \times 4}$.

THE END.

LONDON:—PRINTED BY JAMES MOYES,
Castle Street, Leicester Square.

TRANSCRIBER'S NOTES

1. Silently corrected obvious typographical errors and variations in spelling.
2. Retained archaic, non-standard, and uncertain spellings as printed.

*** END OF THE PROJECT GUTENBERG EBOOK FIRST NOTIONS
OF LOGIC (PREPARATORY TO THE STUDY OF GEOMETRY) ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a)

distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact

links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.